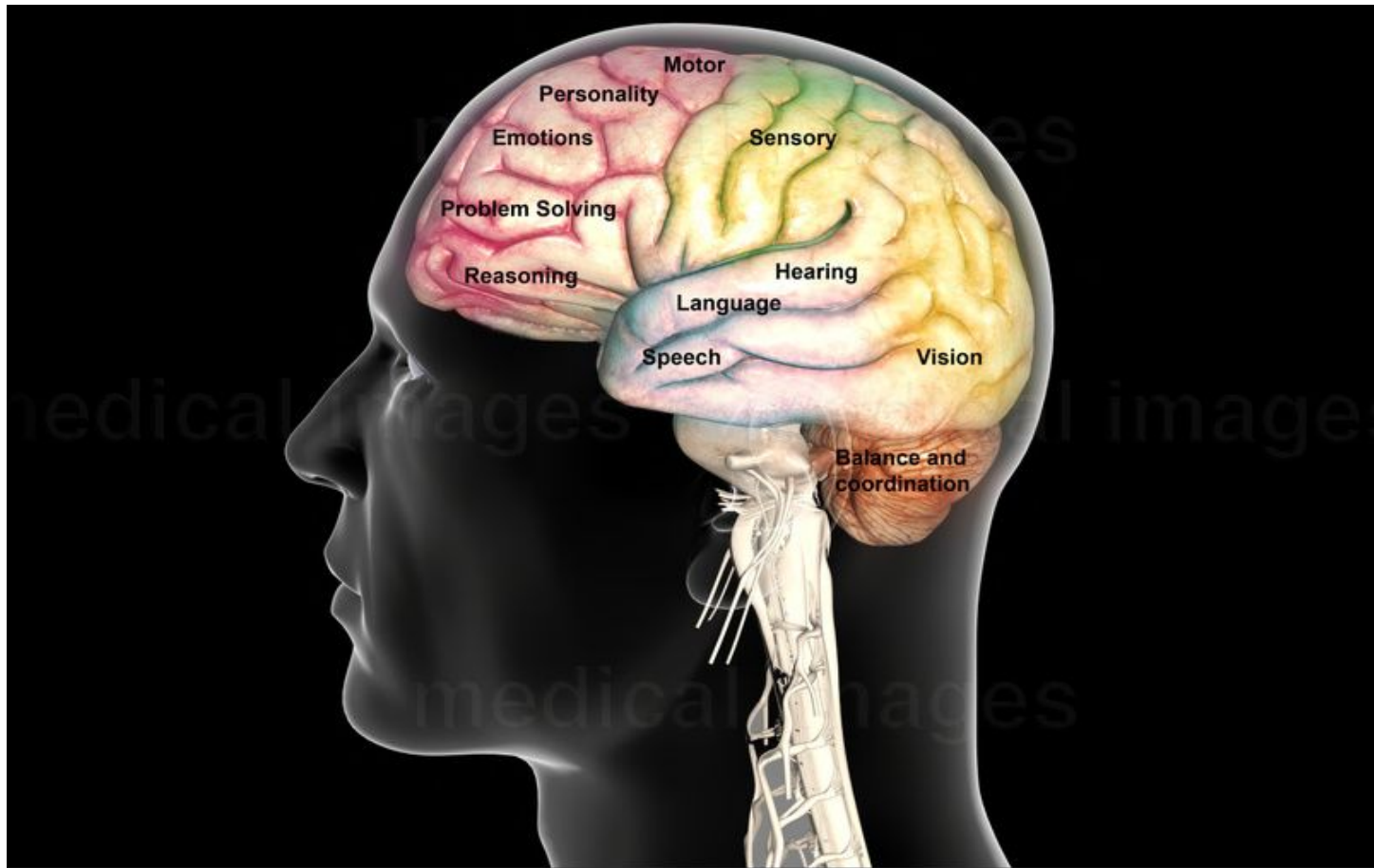


Functional areas of Cerebral Cortex



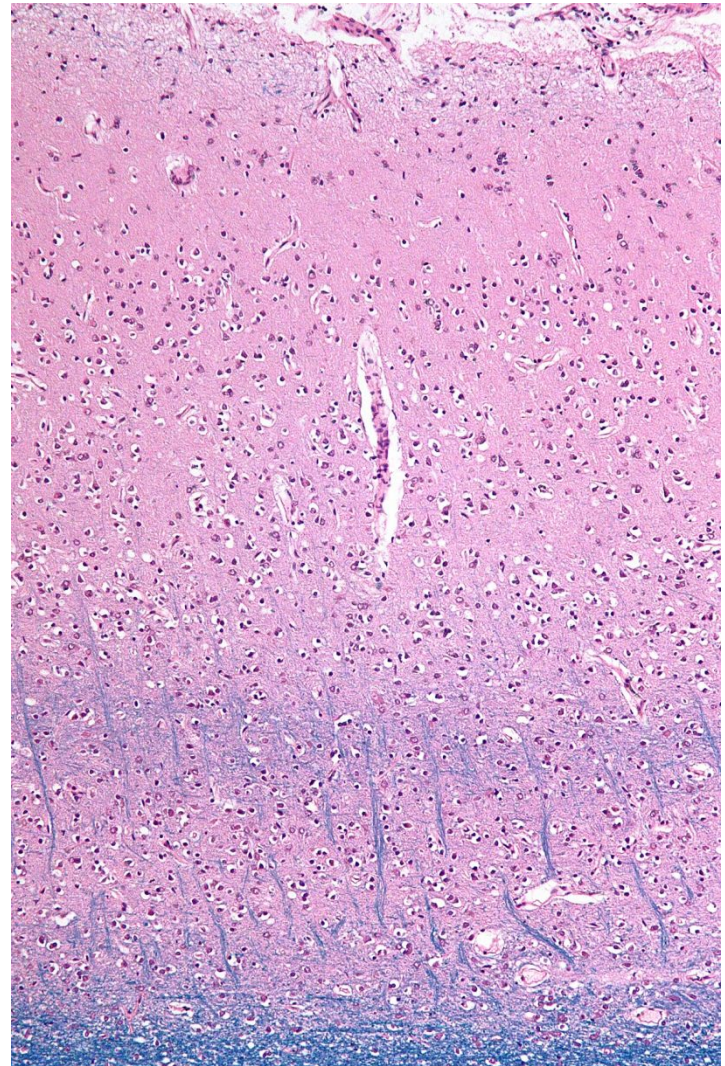
Learning objectives:

At the end of the lecture, the students should be able to define:

- **Primary & secondary motor areas, their main function, & their clinical anatomy.**
- **Primary & secondary sensory areas, their main function, & their clinical anatomy.**
- **Broca & Wernicke's areas & their functions, & their clinical anatomy.**
- **Primary & secondary visual & auditory areas & their functions.**

Histology of cerebral cortex:

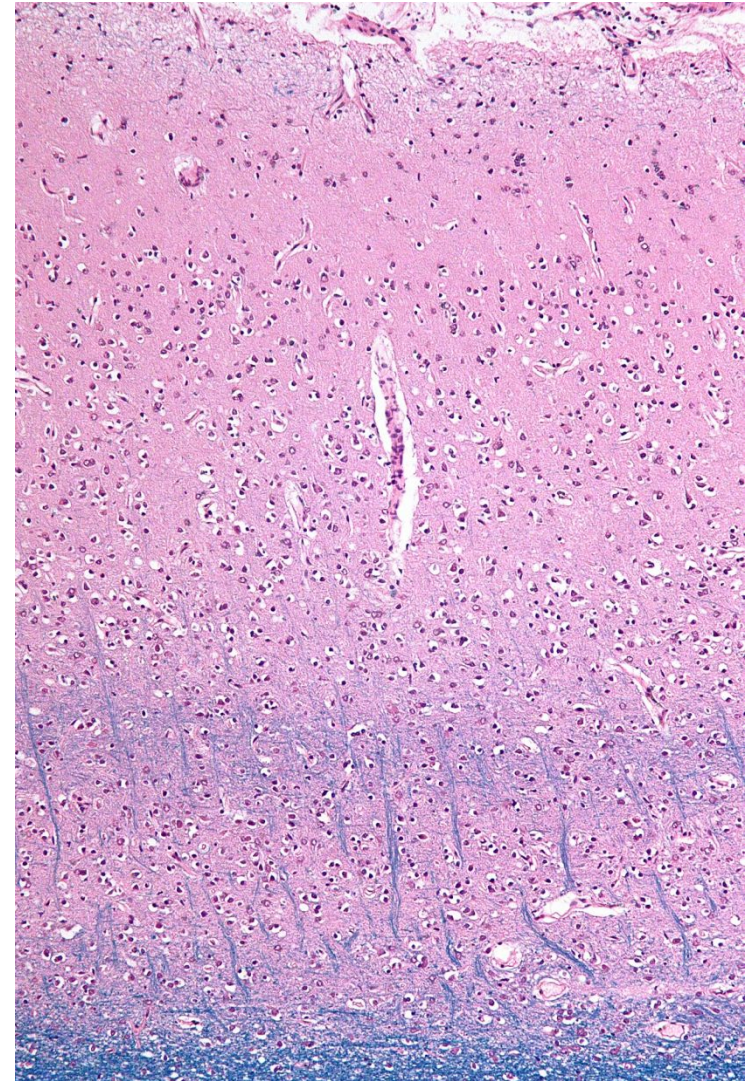
- The cerebrum contains two cerebral hemispheres.
- The outer layer of gray matter covering the cerebral hemispheres is known as ***cerebral cortex*** or ***pallium***.
- The structure of pallium is not same in all parts of cerebrum.
- Around 90% of cerebral cortex has basic **six layers** & is known as ***isocortex***, ***neocortex*** or ***neopallium***.
- **Rest of the 10% cortex is known as allocortex.**



- The areas of cerebral cortex having basic **six layers** are called **homotypical**.
- The allocortex is relatively simpler and contains the olfactory cortex and the cortex covering the hippocampal formation.

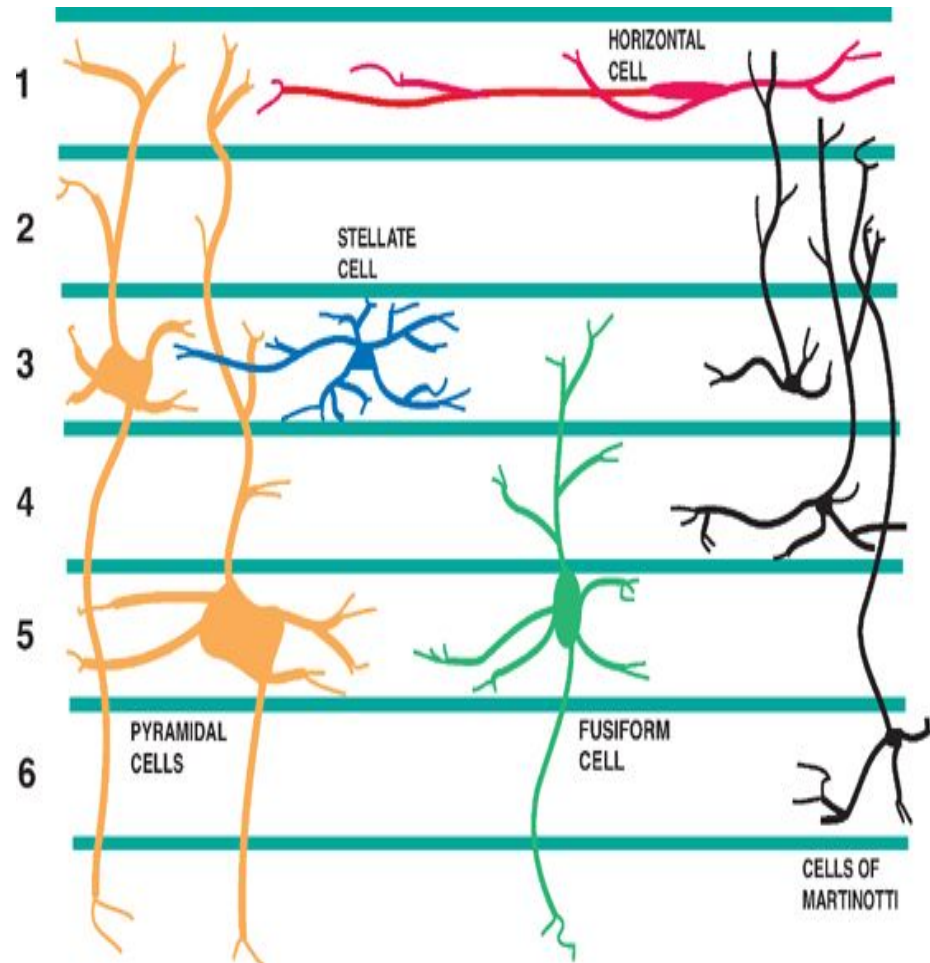
Cells of the cerebral cortex:

- The neurons found in the cerebral cortex can be classified into the two main types:
 - I. **Pyramidal cells**
 - II. **Nonpyramidal cells**



I. Pyramidal cells:

- These are the multipolar neurons with pyramidal cell bodies.
- The length of the cell bodies of these neurons ranges from 10-50 μm .
- Depending on their length they may be classified as small, medium or large.
- Each cell has one main dendrite which arises from the apex of the perikaryon and terminate in the most superficial layer.
- Few other branching dendrites arise from the base.
- The axon arises from the base and continues downward in white matter.



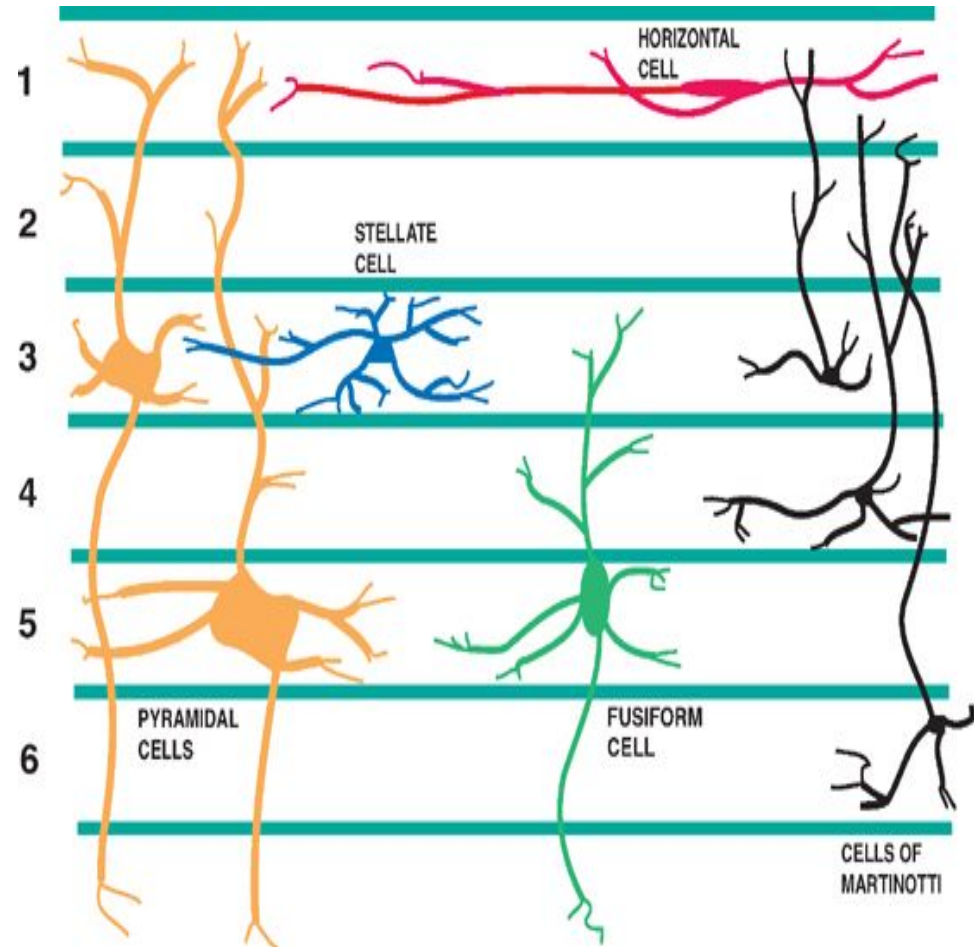
- The giant pyramidal cells in the internal pyramidal layer of primary motor area are called Betz cells having cell bodies up to 100 μm long.

II. **Nonpyramidal cells:**

This group contains the neurons of the following types:

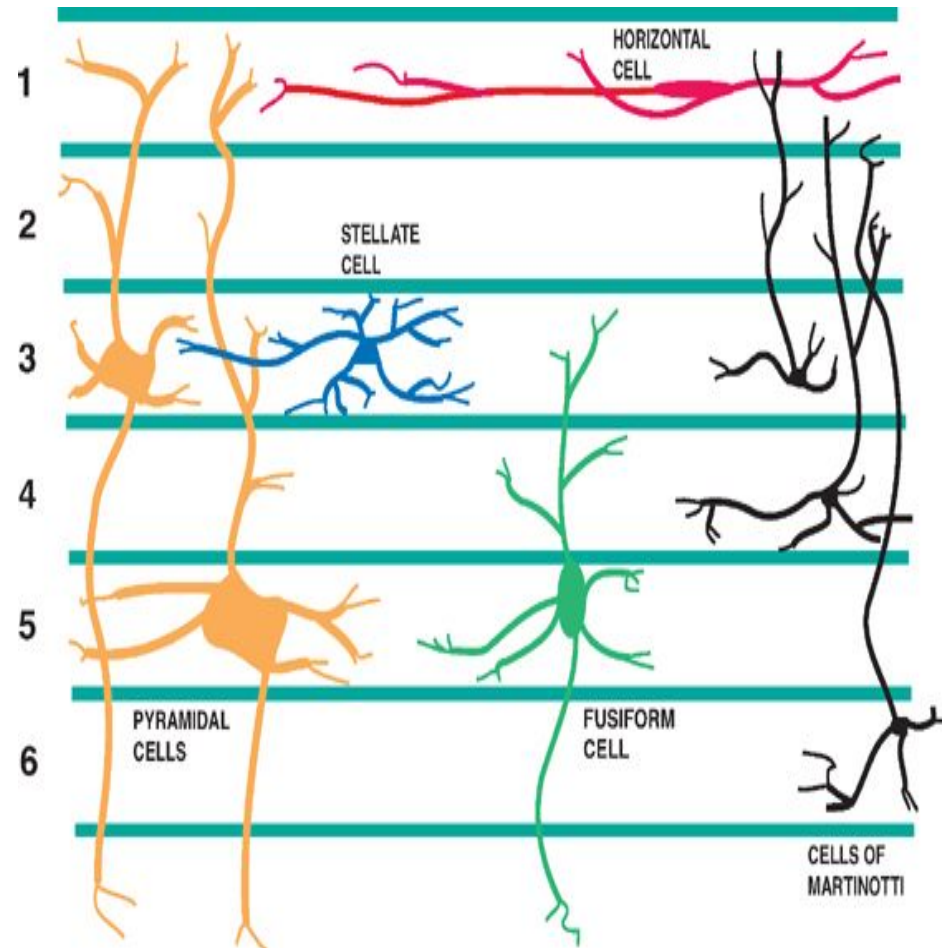
1. **Granule cells:**

- They have small polygonal cell bodies with numerous small dendrites and a short axon.
- The dendrites run in various directions giving them star-shaped appearance, hence called stellate cells.
- These dendrites mostly receive inputs from the related thalamic nuclei.



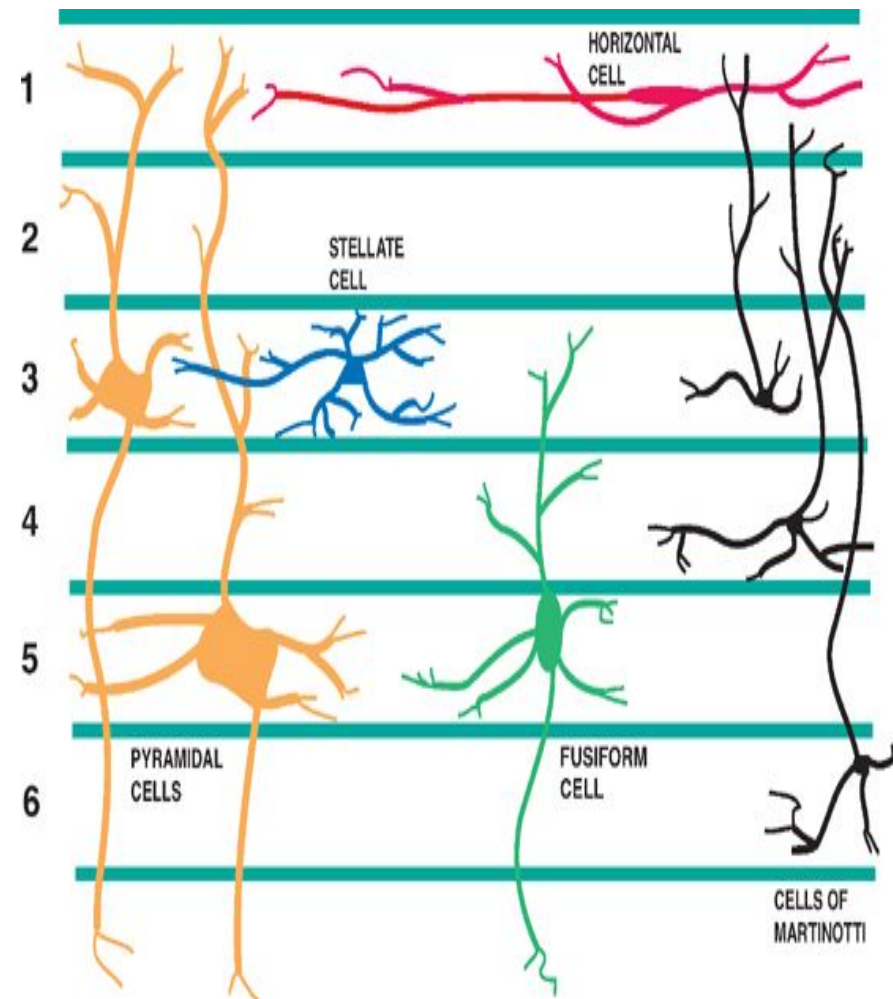
2. Horizontal cells (of Cajal):

- They are small, fusiform and horizontally oriented neurons present in the most superficial cortical layer.
- They have many short dendrites and a long axon that runs parallel to the surface of cortex.
- They synapse with the dendrites of pyramidal cells.



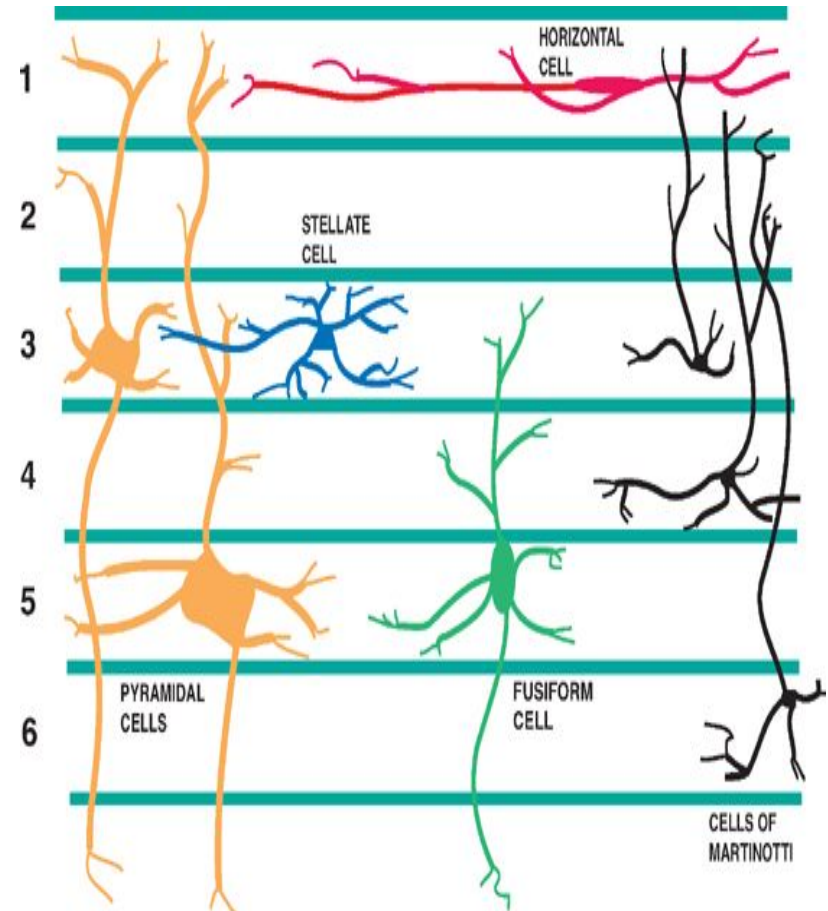
3. Martinotti cells:

- These are small multipolar neurons found in the deeper layers of cerebral cortex.
- They have short dendrites but a long axon that ascends towards superficial layers of cerebral cortex usually ending up in the outermost layer.



4. Fusiform cells:

- They have spindle-shaped cell bodies.
- A single axon arises from the middle of the cell body and descends into the white matter.
- Each end (pole) of the cell body gives rise to branching dendrites that extend to the superficial and deeper layers of the cortex.

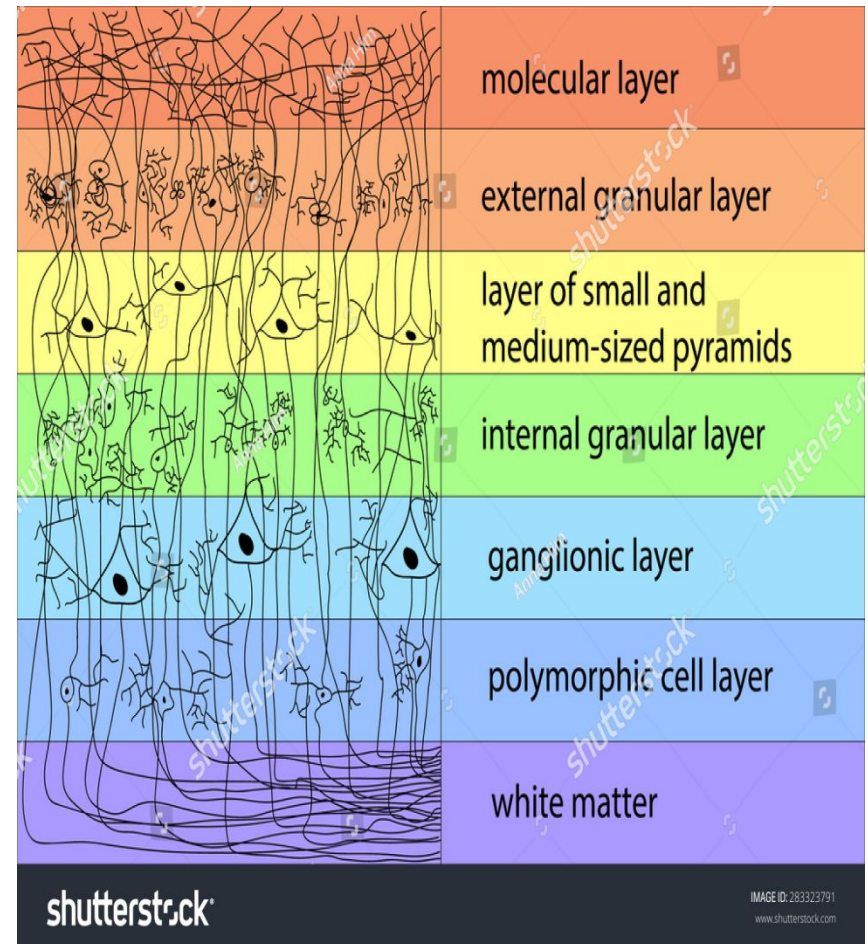


Layers of the cerebral cortex:

- The cerebral cortex (neocortex) consists of six layers of cells which are described below:

1. **Molecular layer:**

- It is a synaptic zone of cerebral cortex.
- It mainly consists of cell processes (dendrites/axons).
- However, some neuroglial & horizontal cells are present.

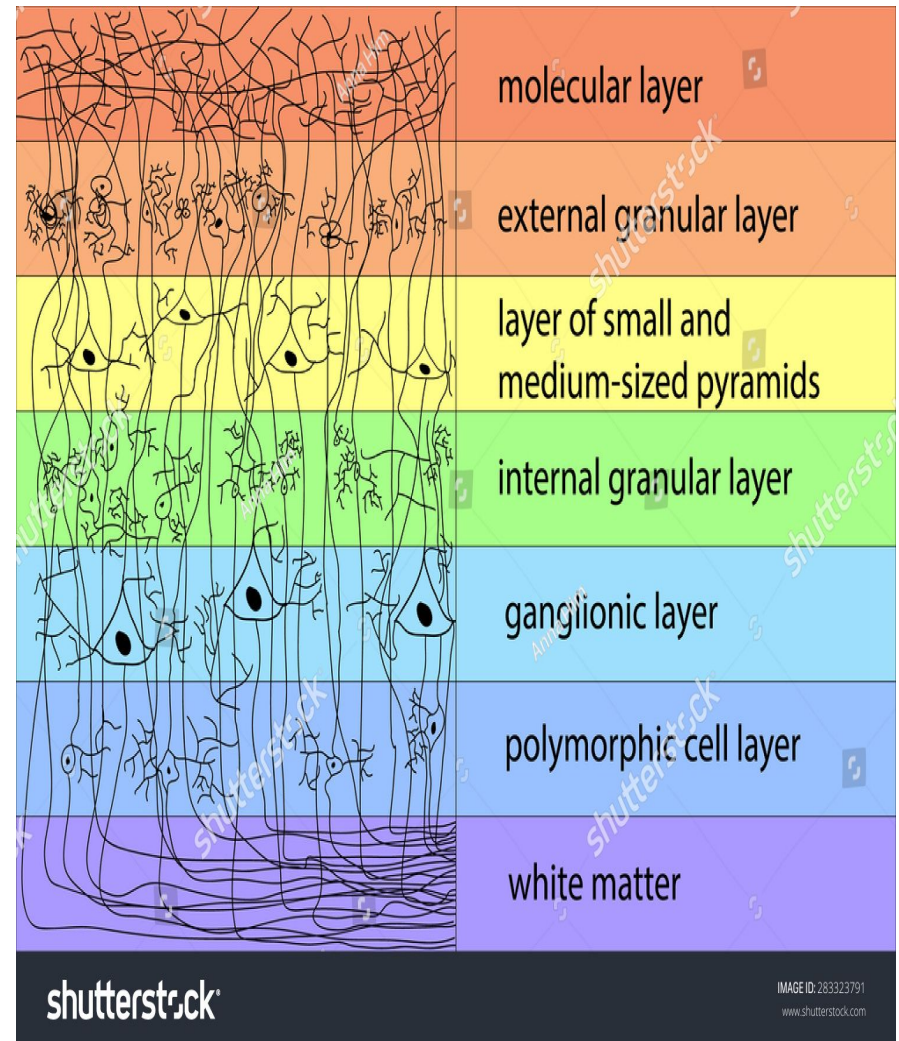


2. External granular layer:

- It mainly consists of closely packed granule cells and many small pyramidal cells.

3. External pyramidal layer:

- It principally comprises of medium pyramidal cells but few granule cells & Martinotti cells are also present.



4. Internal granular layer:

- This also contains closely packed granule cells with many small pyramidal cells.

5. Internal pyramidal layer:

- It contains chiefly large pyramidal cells.
- Some fusiform cells, granule cells, & Martinotti cells are also present.

6. Multiform layer:

- It is the deepest layer, also known as polymorphic layer as it contains neurons of different shapes.
- Hence it acomodates the fusiform cells, pyramidal cells, stellate as well as Martinotti cells.

Allocortex: it consists of only 3 layers: molecular, pyramidal, and multiform layer.

Functional areas of cerebral cortex:

- The three (3) types of functional areas are:

I. **Motor areas:** control voluntary movement.

II. **Sensory areas:** conscious awareness of sensation.

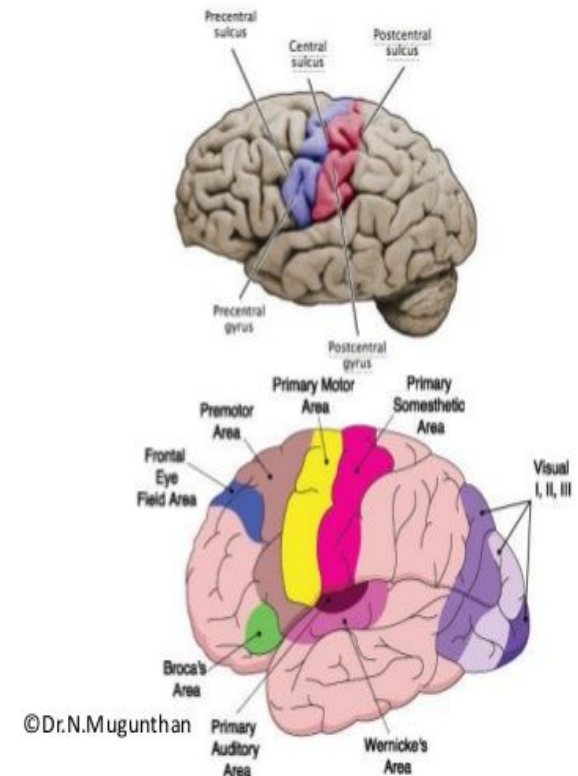
III. **Association areas:** integrate diverse information.

- Conscious behavior involves the entire cortex.

Types of cortical functional areas

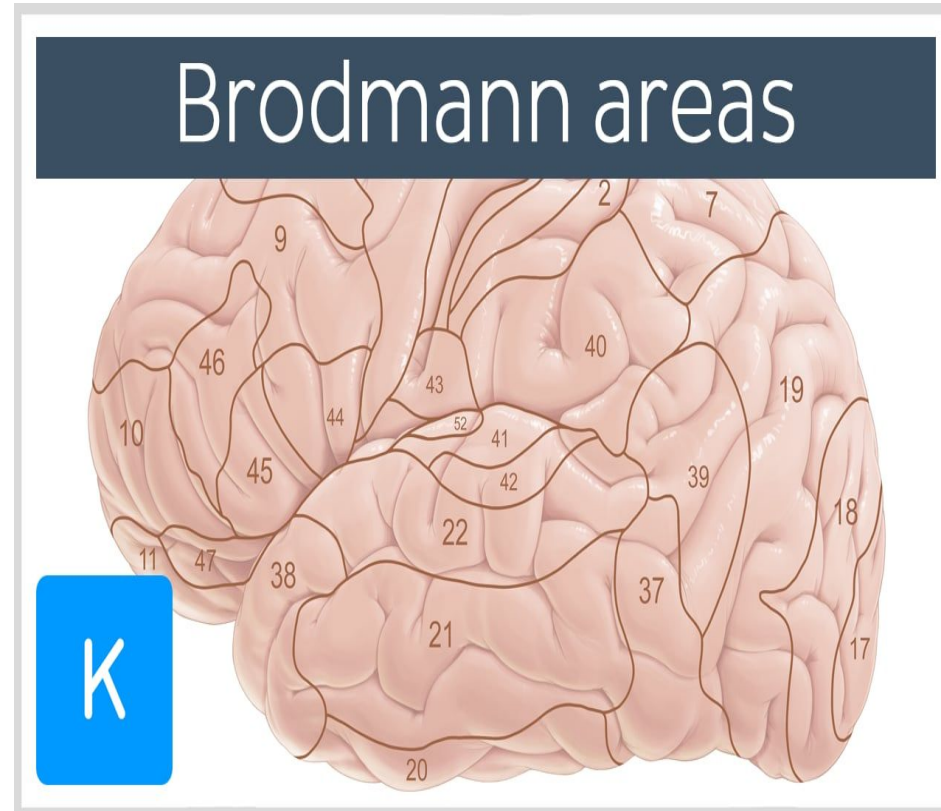
❖ 3 Types of functional areas

1. Motor areas
2. Sensory areas
3. Association areas



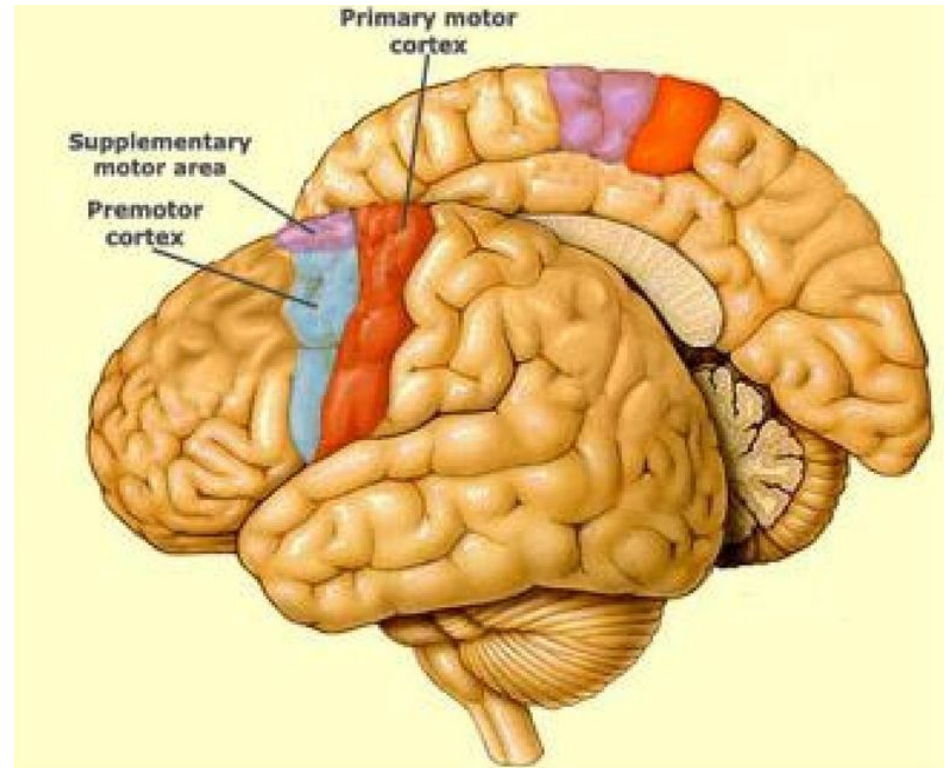
Brodmann's mapping of cerebral cortex:

- A German neurologist, **Korbinian Brodmann** described precise division of cerebral cortex into different areas of specialization.
- **Brodmann mapped cerebral cortex** and defined 52 distinct regions known as '***Brodmann areas***'.
- Initially these areas were described on the basis of cyto-architecture and were later found to be functionally significant.



Frontal lobe:

- **Precentral area** is located in precentral gyrus.
- Histologically, granular layers are absent and pyramidal cells are prominent.
- Most of the corticospinal and corticobulbar fibers arise from pyramidal cells in this area.
- Precentral area is divided into a **posterior area** & an **anterior area**.

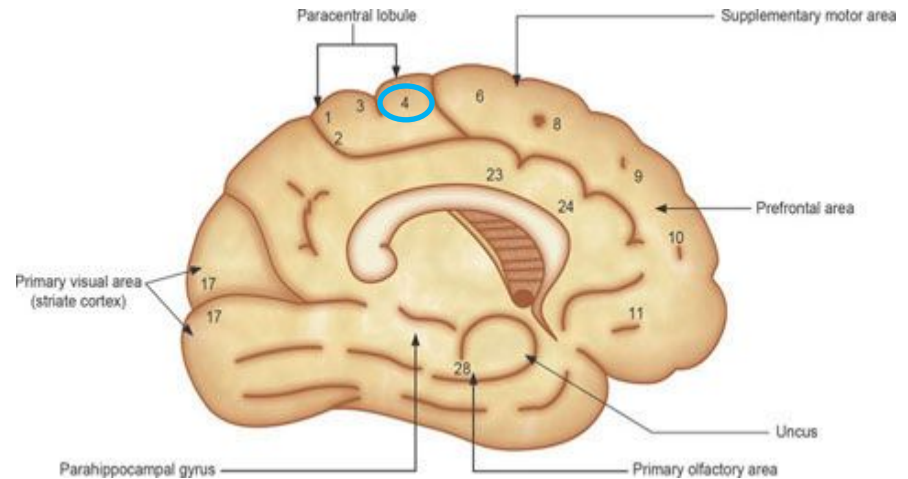
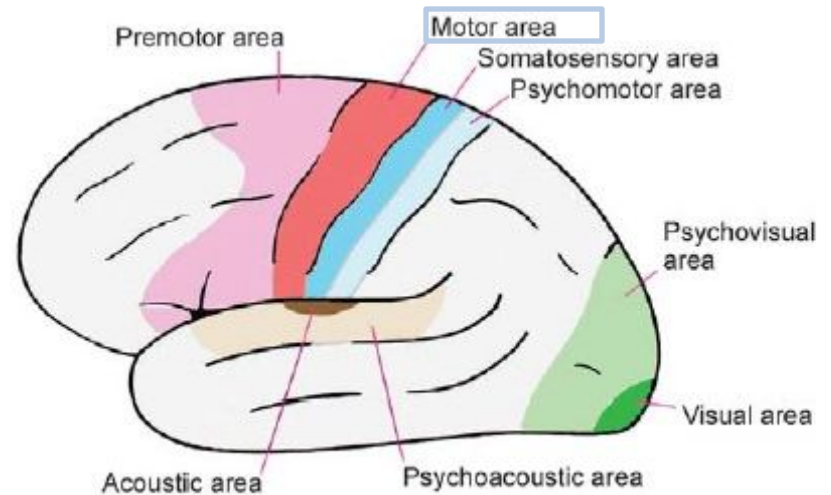


□ **Posterior area:** it is also called **motor area, primary motor area, or Brodmann area 4.**

- It occupies precentral gyrus extending up to paracentral lobule.

Functions:

- To carry out individual movements of different parts of body.
- It produces isolated movements on opposite side of the body and contraction of muscle groups meant to perform a specific movement.

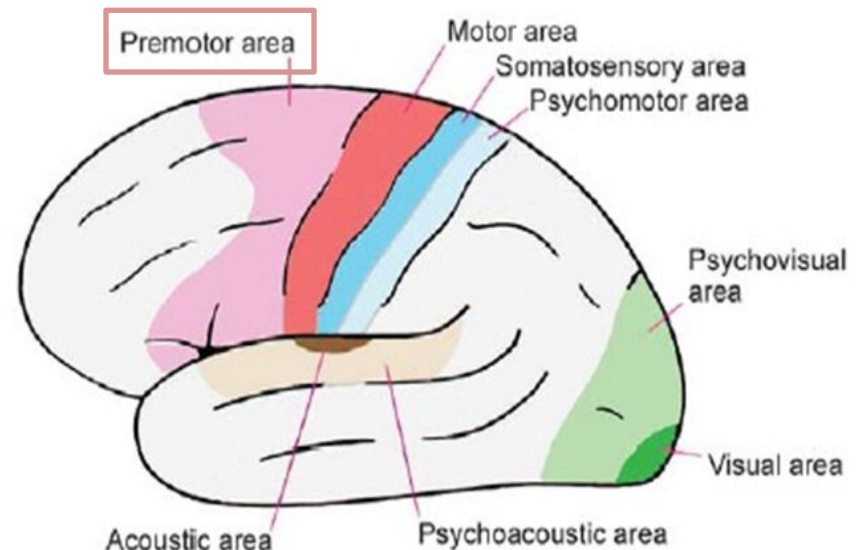


□ **Anterior area:** also known as **premotor area, secondary motor area, or Brodmann area 6.**

- Occupies anterior part of precentral gyrus and posterior parts of superior, middle & inferior frontal gyri.

Functions:

- Receives inputs from sensory cortex, thalamus and basal nuclei.
- Particularly controls coarse postural movements via its connections with basal nuclei.
- Stores programs of motor activity assembled as a result of past experience.



Clinical anatomy:

- Lesions of **primary motor cortex** in one cerebral hemisphere → **paralysis of contralateral extremities** affecting more skilled movements.
- Lesions of **secondary motor area** alone → **impaired skilled movements** with a little loss of strength.
- Destruction of **both the primary & secondary motor areas** → **complete contralateral paralysis**.

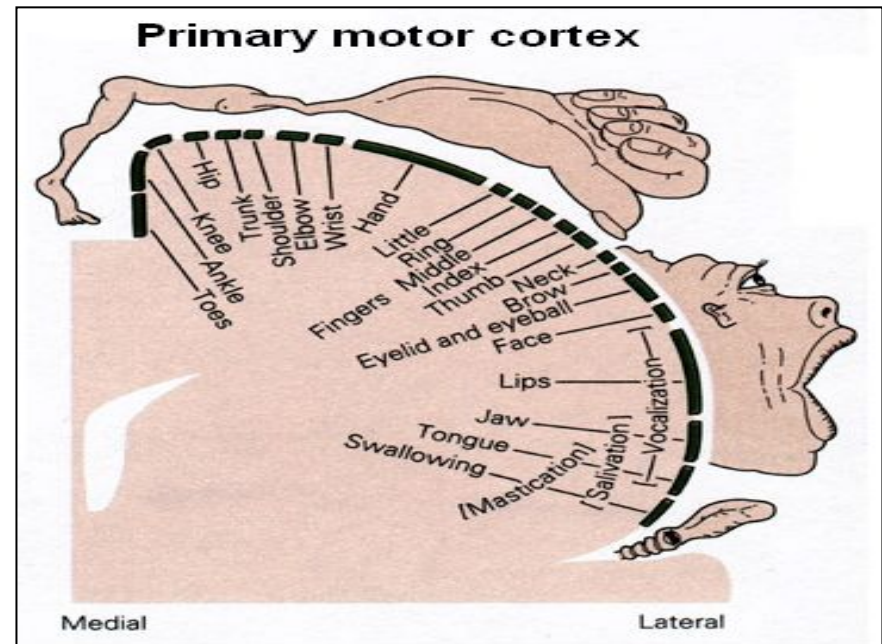
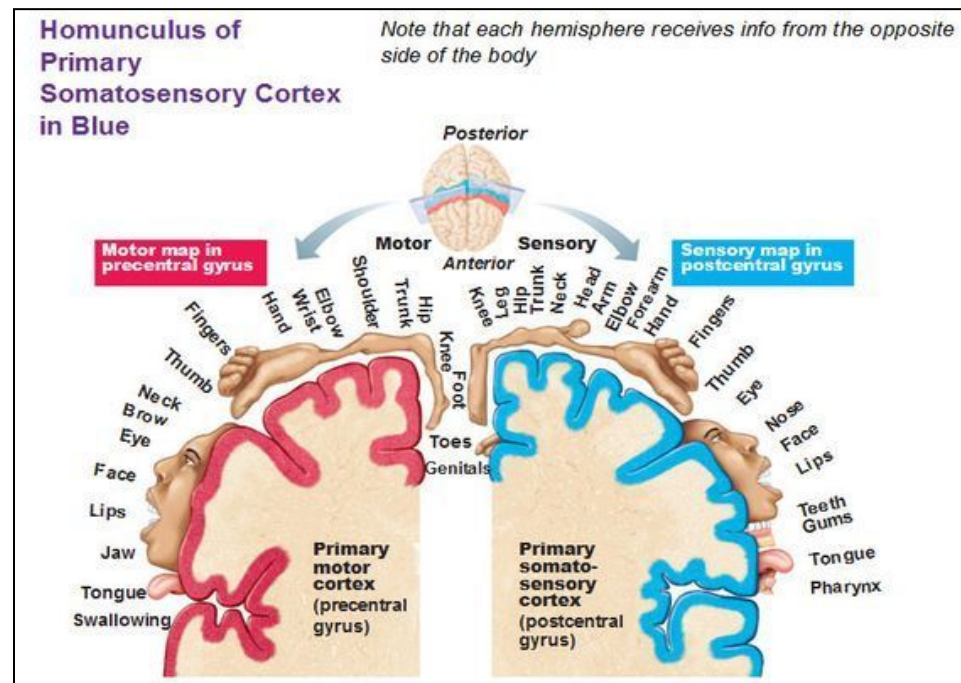
The Homunculus



- Homonculus = “little man” (Latin)
- A **cortical homunculus** is a physical representation of the human body, located within the brain.
- Two types of cortical homonculus:
 - Motor homonculus – pre central gyrus
 - Sensory homonculus – post central gyrus

Motor homunculus on precentral gyrus

- The movement areas in the body are represented in inverted form in precentral gyrus.
- Starting from below and passing superiorly are structures involved in swallowing, the tongue, jaw, lips, larynx, eyelid & brow.
- Proceeding superiorly is an extensive area for movements of fingers like thumb, hand, wrist, elbow, shoulder and trunk.
- Movements of hip, knee and ankle are represented in highest areas of precentral gyrus and toes on medial surface in paracentral lobule.

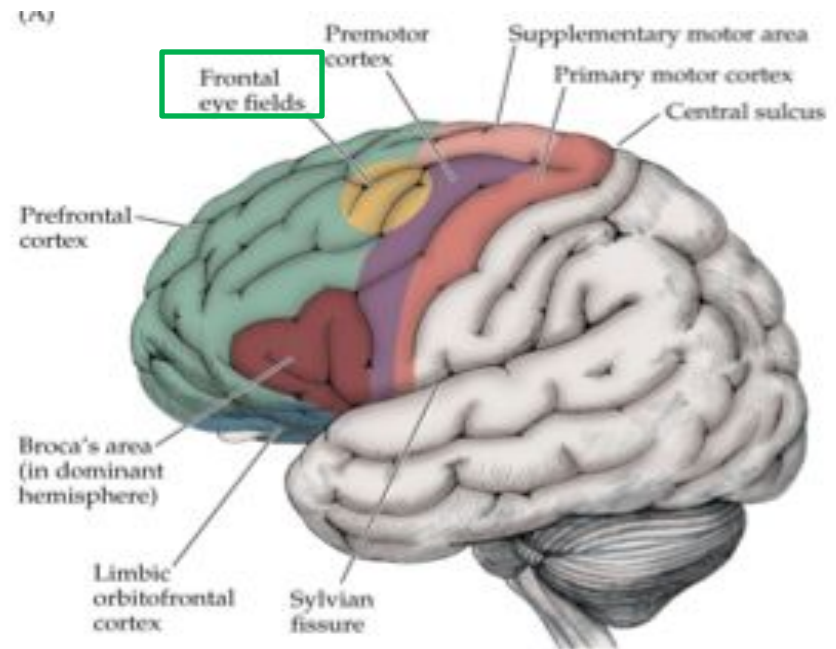


□ **Frontal eye field (Brodmann areas 6,8 & 9):** extends forward from facial area of precentral gyrus to middle frontal gyrus.

- Nerve fibers pass to superior colliculus (midbrain).

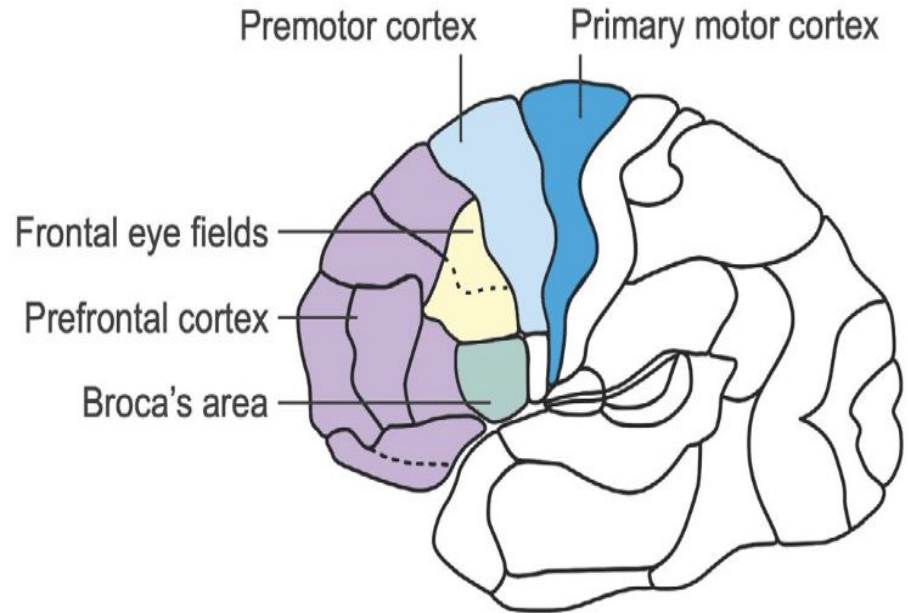
Functions:

- Produces conjugate movement of the eyes, especially towards the opposite side.
- Controls voluntary scanning movements of the eyes.



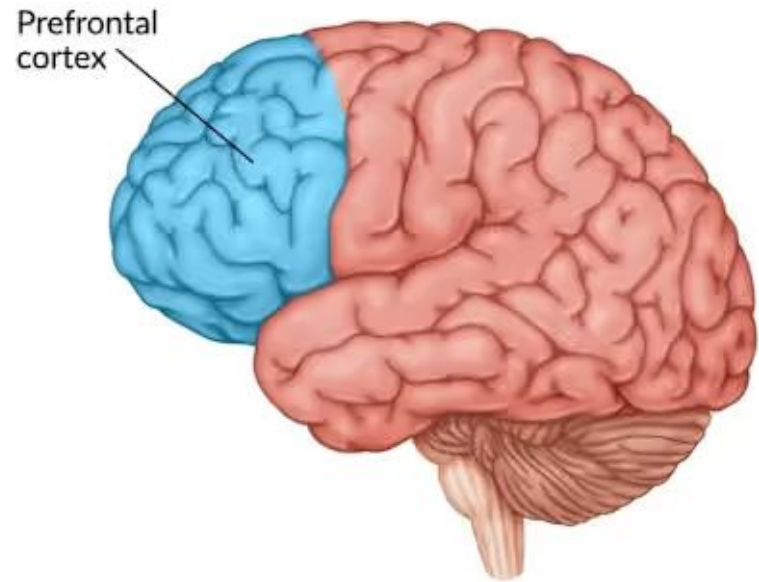
Clinical anatomy:

- In a cerebral hemisphere, the **destructive lesion** of frontal eye field causes **both the eyes deviate to the side of the lesion**.
- Whereas the **irritative lesion** of the frontal eye field of a cerebral hemisphere **deviates both the eyes periodically to the opposite side of the lesion**.



Prefrontal Cortex:

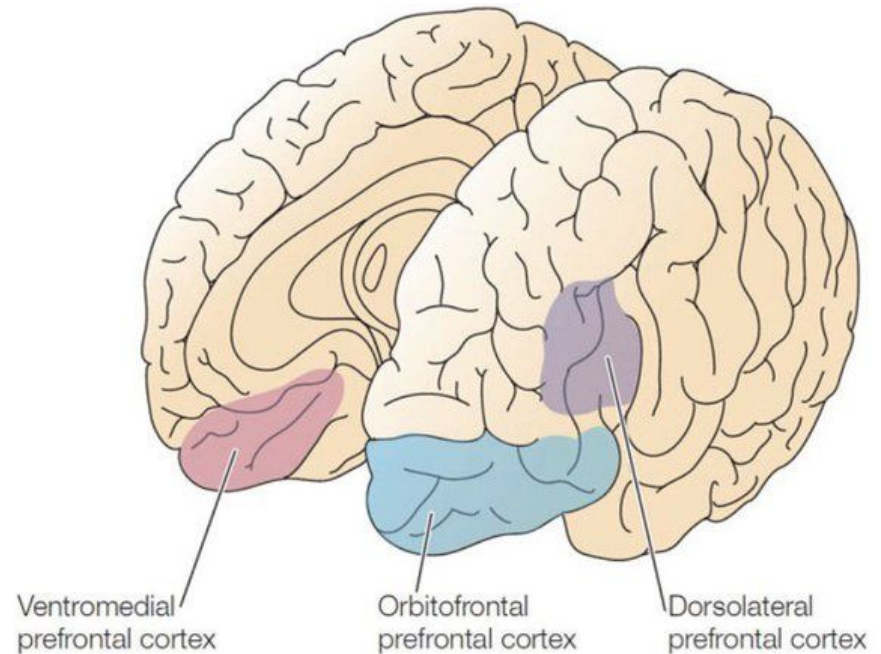
- It is a large area of cerebral cortex that lies anterior to precentral area.
- It includes major parts of superior, middle, & inferior frontal gyri, orbital gyri, most of the medial frontal gyrus, & anterior part of cingulate gyrus (**Brodmann areas 9, 10, 11, & 12**).



- The prefrontal area is related to the **make up of an individual's personality.**
- It is capable of associating experiences needed to create abstract ideas, judgement, emotional regulation, & personality expression.

Clinical anatomy:

- Its damage may cause personality changes, recent memory loss, poor decision-making skills, decreased attention, & stubbornness etc.



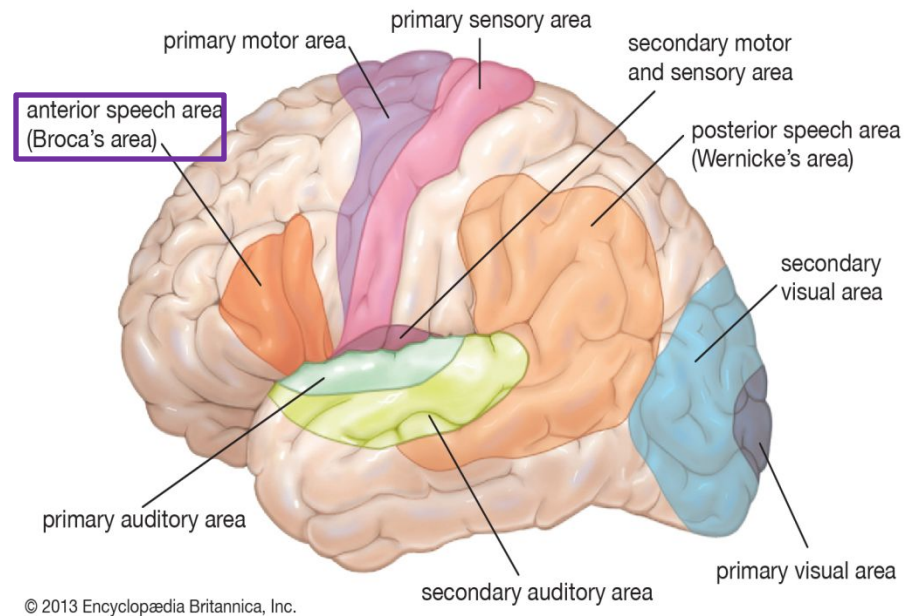
□ **Motor speech area of Broca (Brodmann areas 44 & 45):**
located in the inferior frontal gyrus.

Functions:

- Brings about the formation of words i.e. understandable speech.

Clinical anatomy:

- Destructive lesions in the left inferior frontal gyrus → expressive aphasia (inability to produce speech).



Parietal lobe:

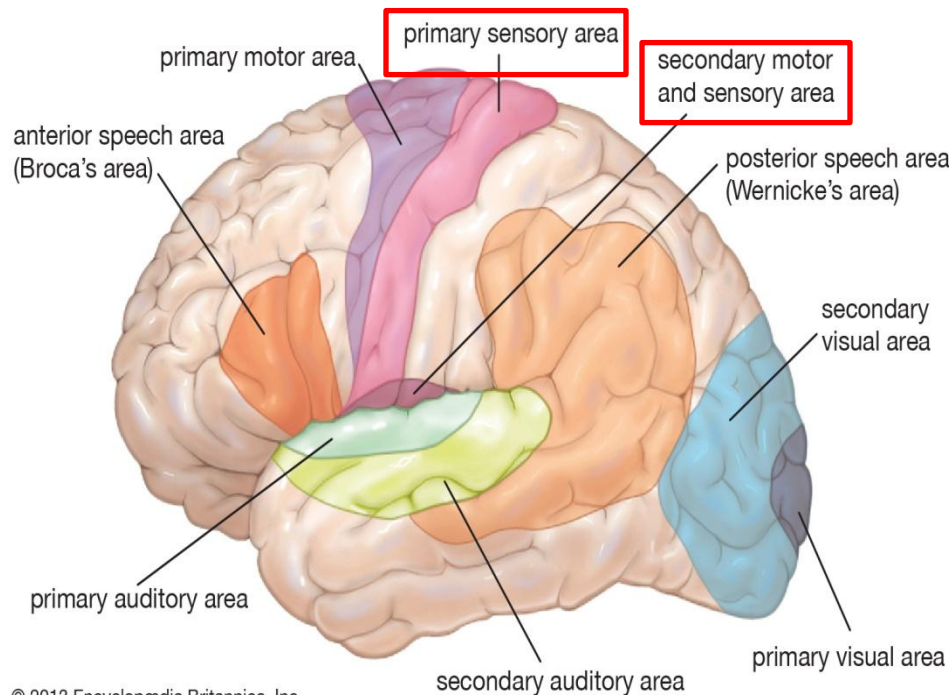
□ **Primary somesthetic area or primary somatic sensory cortex-S1 (Brodmann areas 3,1 & 2):**

occupies postcentral gyrus and posterior part of paracentral lobule on medial surface.

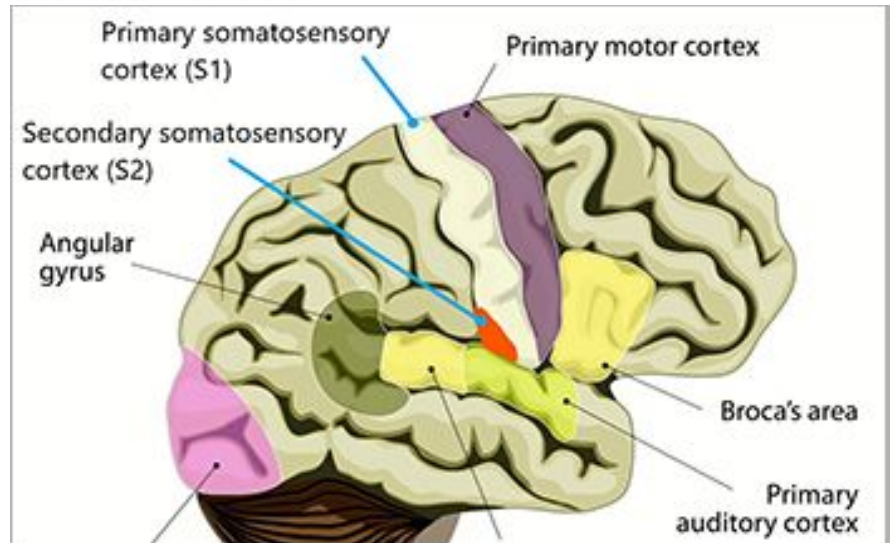
- Receives projection fibers from ventral postero-lateral and ventral postero-medial nuclei of thalamus, and, also
- Receives afferent fibers from muscle spindles, tendon organs and joint receptors.

Functions:

- Receives sensations from contralateral side of the body and same side from oral region and both sides from pharynx, larynx etc.



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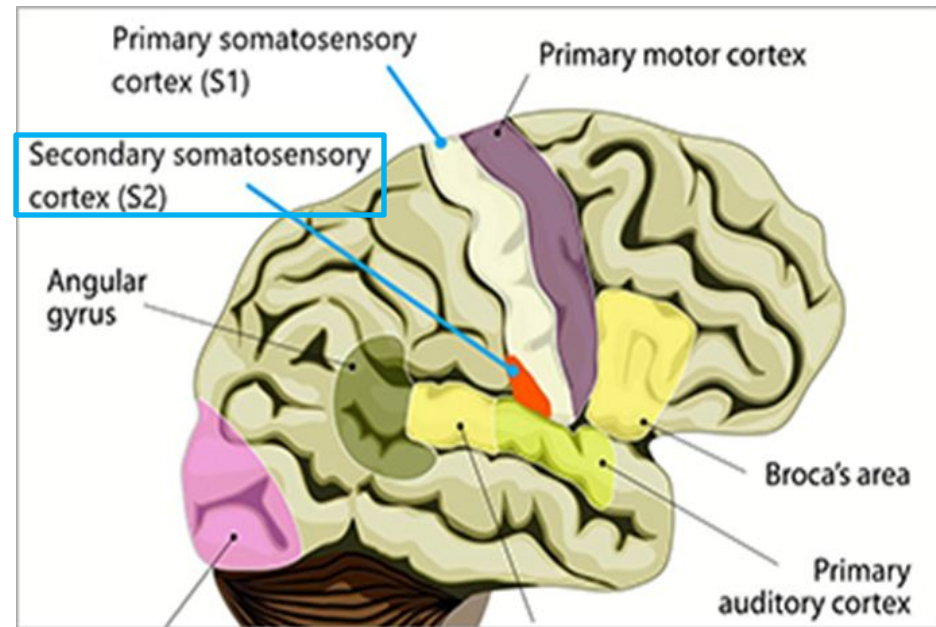


□ **Secondary somesthetic area** or **secondary somatic sensory cortex S2**: located above the posterior limb of lateral fissure.

- Receives impulses from primary sensory cortex and brainstem.

Functions:

- Respond to transient cutaneous stimuli like tapping of skin or brush strokes.

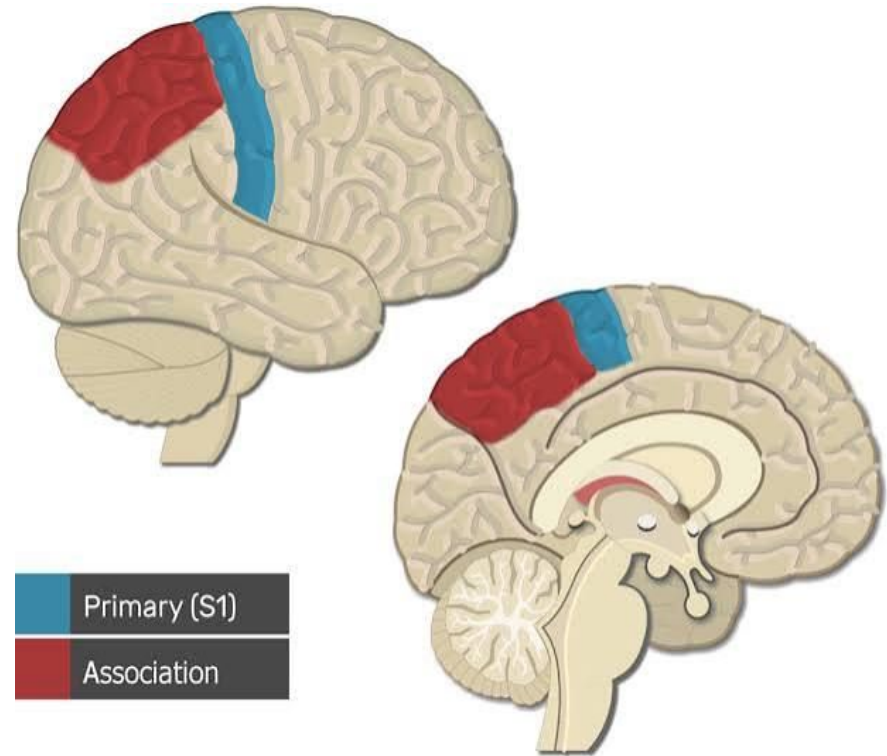


Somesthetic Association Area (Brodmann areas 5 & 7):

- It occupies superior parietal lobule (extending upto the medial aspect) of cerebral hemisphere.
- It has many connections with the other sensory areas of cortex.

Function:

- It receives & integrates different sensory modalities & makes one able to recognize the objects placed in the hand, without the help of vision (**stereognosis**).

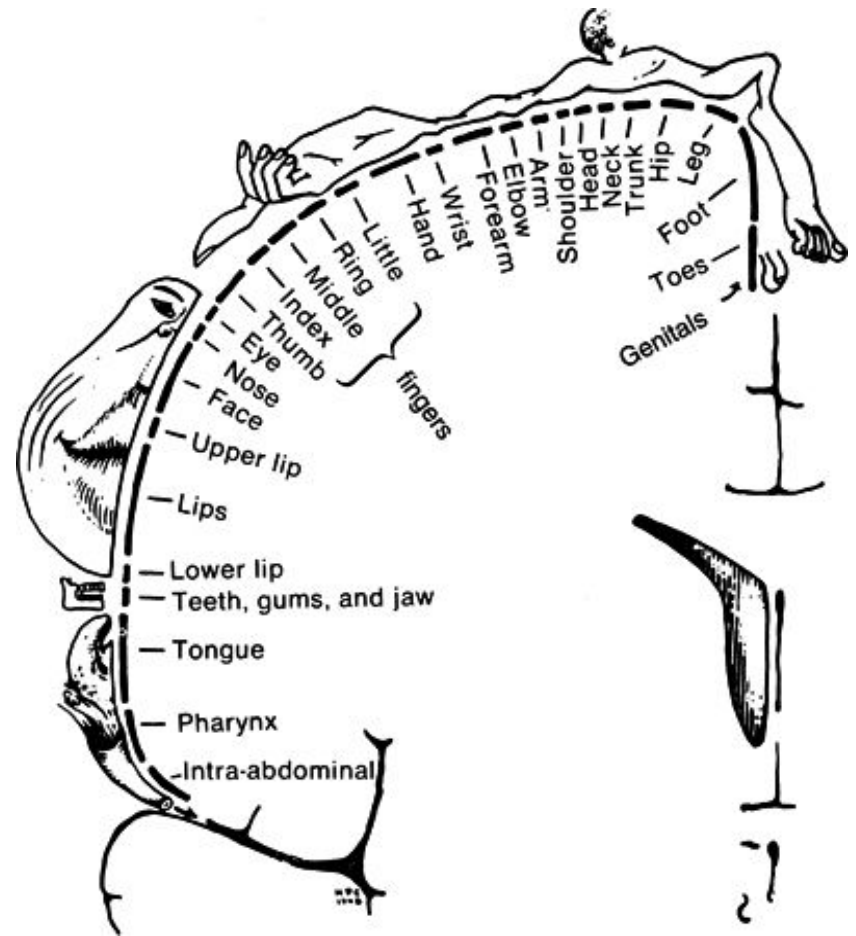


Clinical anatomy:

- Lesions of **primary somesthetic area** of the cortex may lead to **contralateral sensory disturbances**, affecting more the distal parts (of the limbs).
- The patient becomes unable to feel the intensity of warmth, to locate tactile stimuli exactly & to judge weight of the objects.
- Lesions of **somesthetic association area** (superior parietal lobule) may result in **astereognosis**, i.e, loss of integration of sensory impulses.
- An individual is unable to identify objects through tactile exploration, without visual input.
- For example, an individual with closed eyes, will not recognize a key placed in his hand.
- The lesions of **secondary somesthetic area** of the cortex does not cause recognizable sensory defects.

Sensory homunculus on postcentral gyrus

- The opposite half of the body is represented as inverted.
- The pharynx, tongue and jaws are represented in the most lower part of postcentral gyrus.
- Then proceeding higher face, fingers, hand, arm, trunk and thigh.
- Leg, foot, anal and genital areas are found on medial surface in paracentral lobule.
- Size of cortical areas allocated to each part of the body is directly proportional to the number of sensory receptors in that part of body.



Occipital lobe:

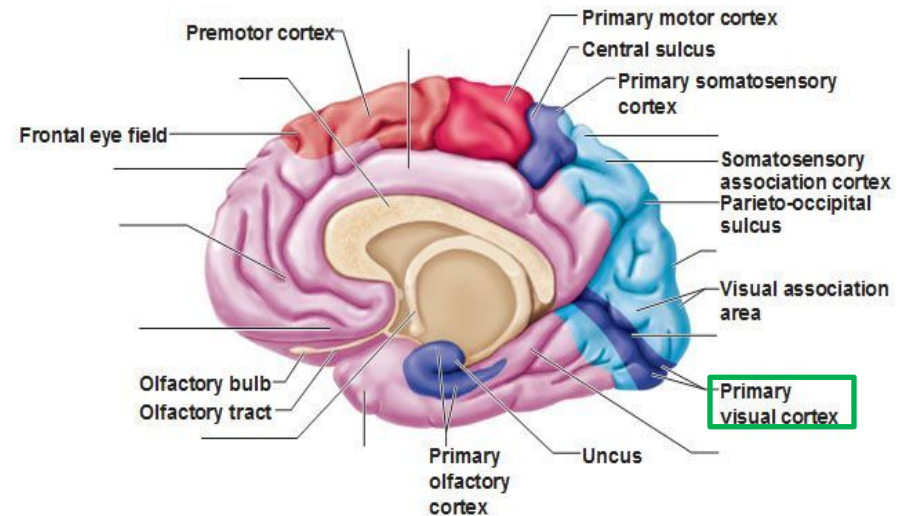
□ **Primary visual area (Brodmann area 17):** is situated in the posterior part of calcarine sulcus.

Function:

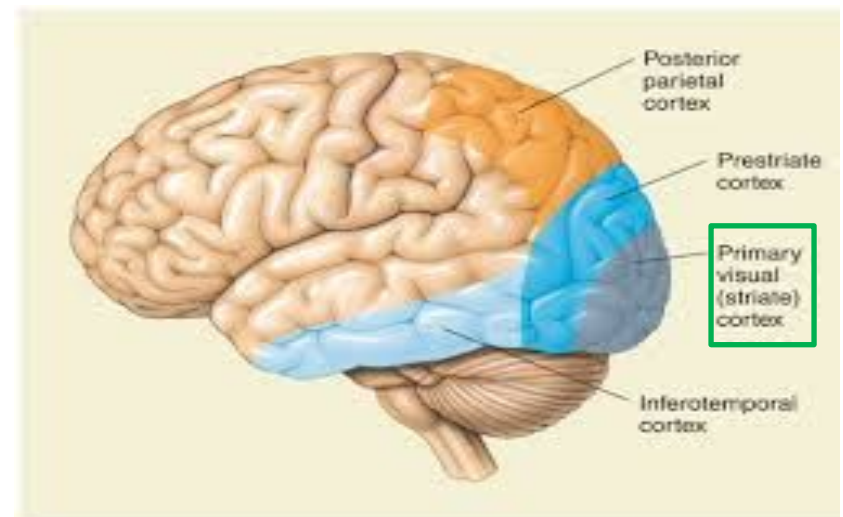
- The macula lutea or central area of retina is represented in posterior part of area 17 of cortex.
- The visual impulses from peripheral parts of retina terminate in the anterior part of area 17.

Functional Areas of the Cerebral Cortex

(b) Parasagittal view, right hemisphere



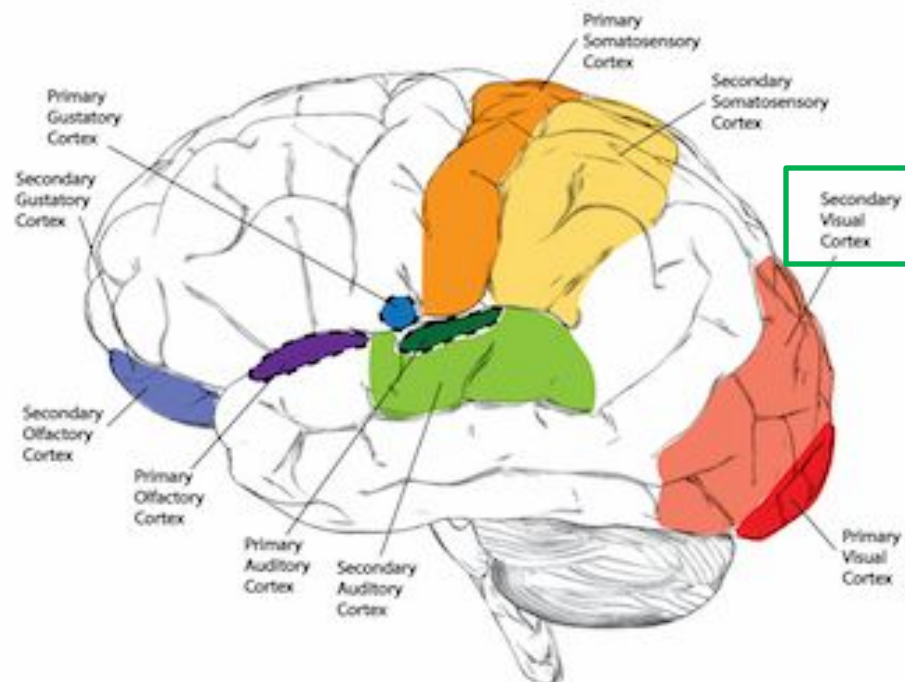
- | | |
|-------------------------------|----------------------------|
| Primary motor cortex | Motor association cortex |
| Primary sensory cortex | Sensory association cortex |
| Multimodal association cortex | |



- **Secondary visual area (Brodmann areas 18 & 19):** surrounds primary visual area on medial and lateral surfaces of cerebral hemisphere.
- Receives afferent fibers from area 17 and other cortical areas and thalamus.

Function:

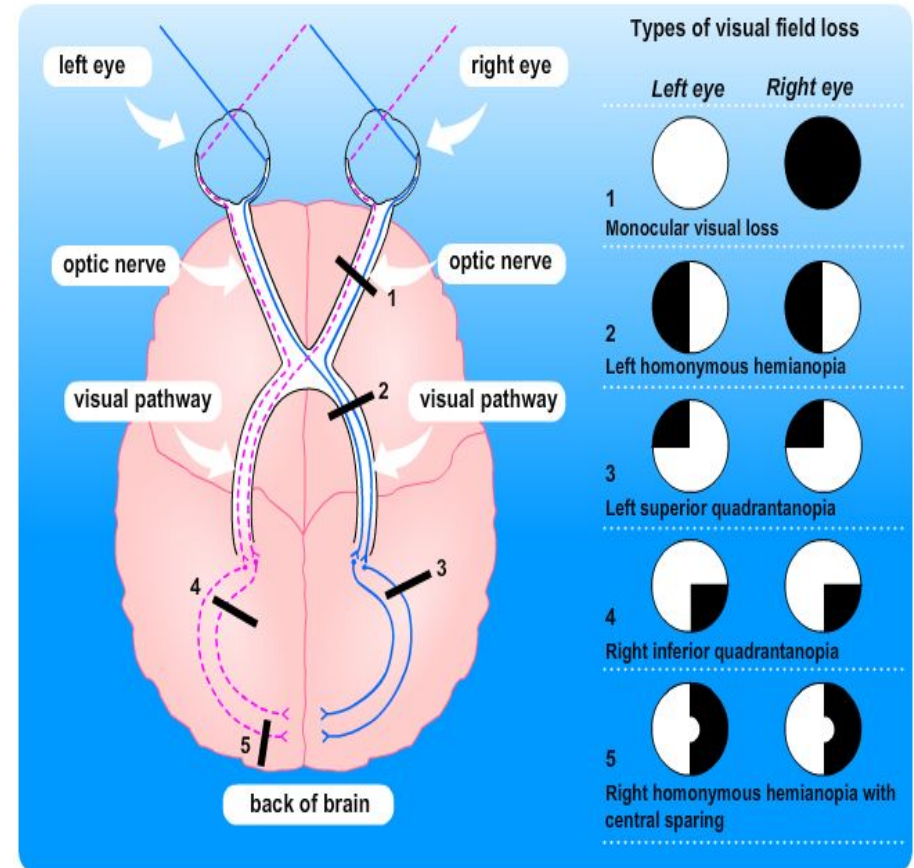
- Relates visual information received by primary visual area to past visual experiences, hence enables the individual to recognize what he/she is seeing.



Clinical Anatomy

(Primary visual area):

- Lesions in the walls of **posterior part of one calcarine sulcus** result in **crossed homonymous hemianopia** i.e, loss of sight in the opposite visual field.
- In the above case central part of visual field is normal due to **macular sparing**.
- Lesions in the **area above calcarine sulcus** result in **inferior quadrantic hemianopia**.
- Lesions in the **area below calcarine sulcus** result in **superior quadrantic hemianopia**.



Temporal lobe:

- **Primary auditory area (Brodmann areas 41 & 42):** includes gyrus of Heschl (transverse temporal gyri) and is situated in the inferior wall of lateral sulcus.

Function:

- Reception of low frequency sounds (anterior part) and high frequency sounds (posterior part).

Functional Areas of the Cerebral Cortex

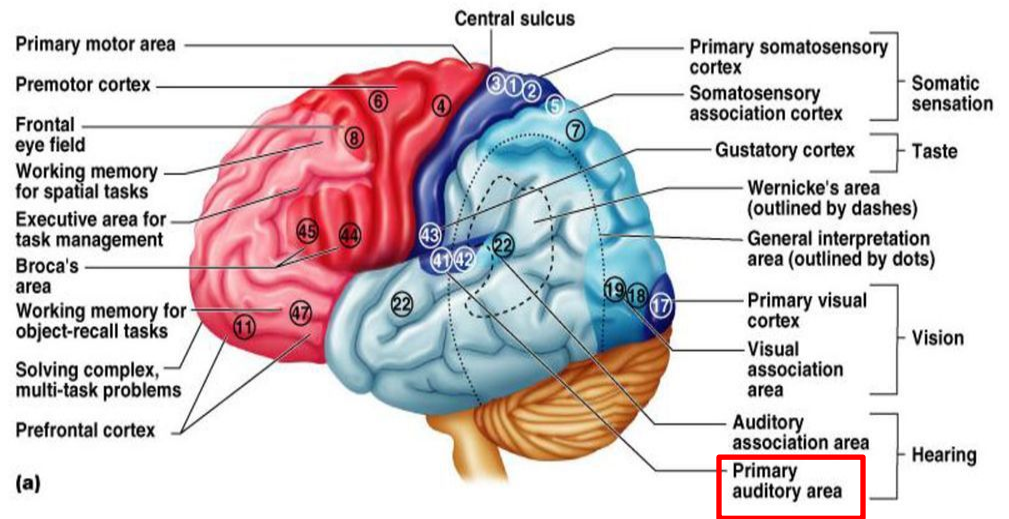


Figure 12.8a

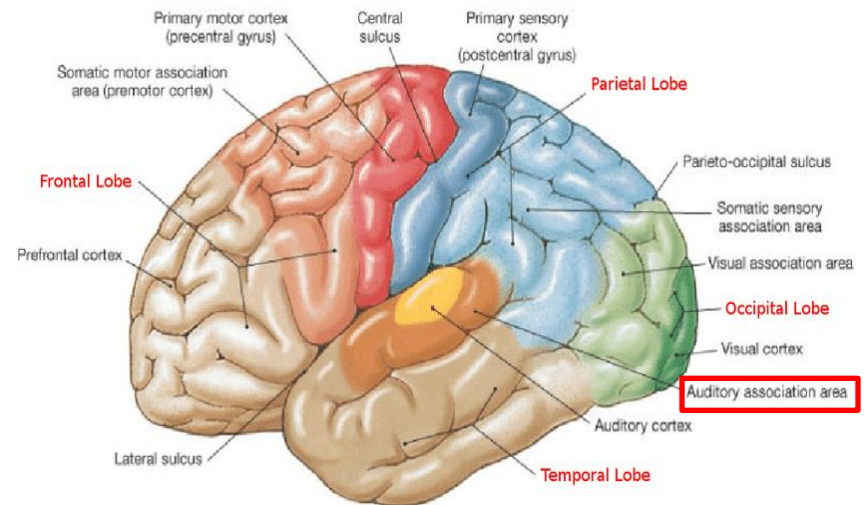
□ Secondary auditory area or auditory association area:

situated posterior to primary auditory area in lateral sulcus and in superior temporal gyrus (**Brodmann area 22**).

- Receives impulses from primary auditory area and thalamus.

Function:

- Interpretation of sounds and association of auditory input with other sensory information.



Clinical anatomy:

Primary auditory area:

- Lesion in one primary auditory cortex will result in bilateral loss of hearing, more in the opposite ear.
- It causes loss of ability to locate the sounds.
- Bilateral destruction of primary auditory cortex ends up in total deafness.

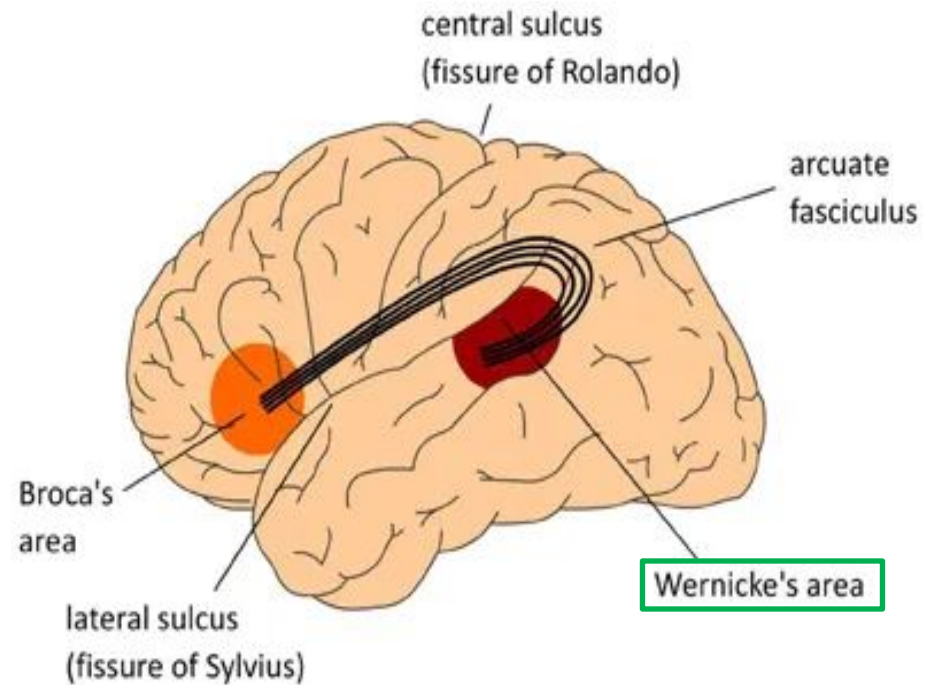
Secondary auditory area:

- Lesions of the cortex posterior to primary auditory cortex in lateral sulcus & superior temporal gyrus result in **acoustic verbal agnosia (word deafness)** i.e, inability to interpret sounds.

- **Sensory speech area of Wernicke:** localized in mainly in superior temporal gyrus but extended around posterior end of lateral sulcus in parietal region.
- Wernicke area is connected to the Broca area by the **arcuate fasciculus**.

Function:

- Allows the understanding of written and spoken language and enables a person to read a sentence, understand it and say it out loud.



Clinical anatomy of Sensory Speech Area of Wernicke:

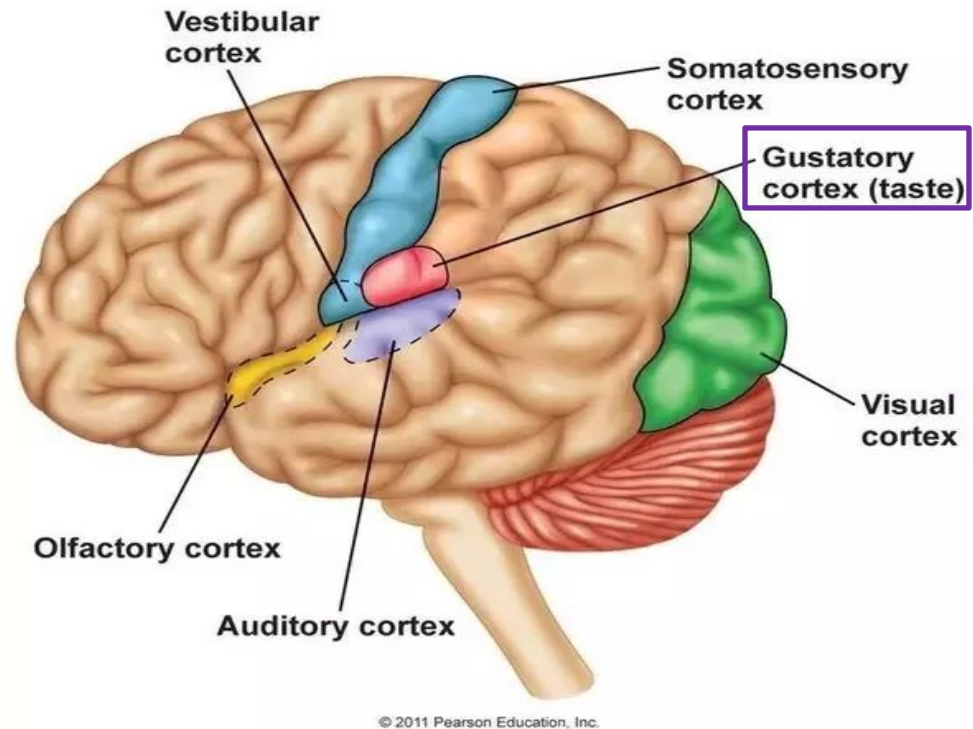
- Destructive lesions of **Wernicke's area** in the dominant hemisphere → loss of ability to understand the spoken and written word, called **receptive aphasia**.

Clinical anatomy of the Motor & Sensory Speech Areas:

- Destructive lesions involving both the **Broca & Wernicke speech areas** → loss of the production of speech and the understanding of spoken and written word is called **global aphasia**.

Other cortical areas:

□ **Taste area:** situated at the lower end of postcentral gyrus in superior wall of lateral sulcus and adjoining area of insula (Brodmann area 43).



Dominant Angular Gyrus:

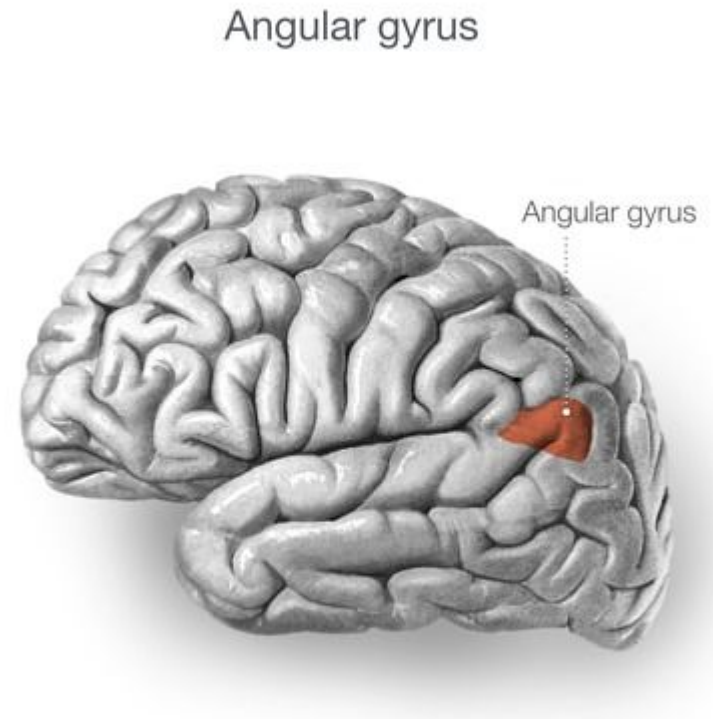
- This is considered as a part of Wernicke area in the posterior part of inferior parietal lobule (**Brodmann area 39**).

Function:

- It is associated with complex language related functions such as reading, writing) to make sense of the meaning and content of written words.

Clinical anatomy:

- Its lesions may cause **alexia (inability to read) & agraphia (inability to write)**.



References:

- **CLINICAL NEUROANATOMY**

Richard S. Snell

8th edition

Chapter 8

Pages # 288 - 294